ANATOMY

Pelvis & Perineum

LG 1 · Dr. Nali Abdulkader Maaruf
GUE BLOCK

Anatomy GUE Block · MEDucation by Mattin Majid

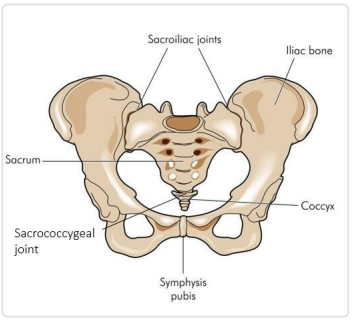
BEFORE YOU READ — ANSWER FROM MEMORY

1. What forms the pelvic inlet anteriorly, posteriorly and laterally?
2. Which gland lies in the superficial perineal pouch of a woman — and which gland sits in the deep pouch of a man?
3. Name the medial and lateral walls of the ischiorectal fossa, and the canal that runs in it.

Guessing wrong before reading still improves retention — attempt all three, then check as you go.

1 Pelvis and bony pelvis

Pelvis	Lower part of the trunk **between the abdomen and the lower limb** · has the bony pelvis and a cavity
Perineum	Region **below the pelvic diaphragm**, between the thighs · contains the external genitalia
Bones (4)	**Two hip bones** + **sacrum** + **coccyx**
Job	A **strong, stable link** between the trunk and the lower limb
Joints	**Symphysis pubis** (hip bones meet in front) · **sacroiliac** joints (hip bone ↔ sacrum, behind) · **sacrococcygeal** joint

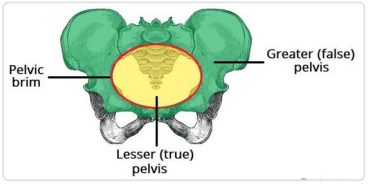


Sacroiliac joints, sacrococcygeal joint, symphysis pubis.

2 Greater vs lesser pelvis — and the two apertures

	False (greater, major) pelvis	True (lesser, minor) pelvis
Where	**Above** the pelvic brim	**Below** the pelvic brim / inlet
What it is	**Part of the abdominal cavity**	The **true pelvic cavity**

	Pelvic inlet (= brim)	Pelvic outlet
Other name	**Superior** pelvic aperture	**Inferior** pelvic aperture · **closed by the pelvic diaphragm**
Anterior	Symphysis pubis	Symphysis pubis
Posterior	**Promontory** and ala of sacrum	**Coccyx**
Lateral	**Iliopectineal (arcuate) lines**	Anterolateral: **ischiopubic ramus** · posterolateral: **sacrotuberous ligament**

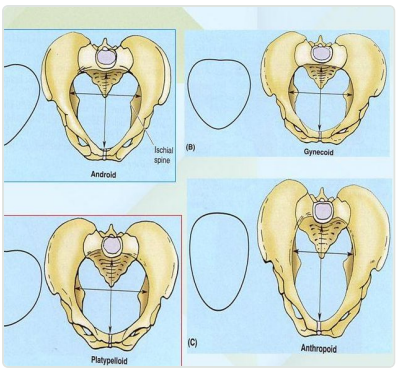


Pelvic brim divides greater (false) from lesser (true) pelvis.

3 Types of female bony pelvis

The female pelvis matters for **obstetrics**: it is the bony canal the baby passes through at birth.

Gynecoid	**Classical female** type · most favourable for childbirth
Android	**Resembles the typical male** pelvis
Anthropoid	**Anteroposterior > transverse** diameter
Platypelloid	**Transverse > anteroposterior** diameter

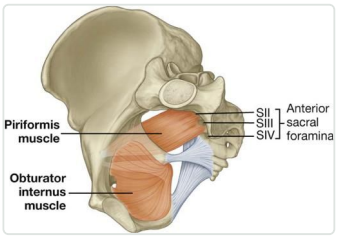


Android, gynecoid, platypelloid, anthropoid inlets.

Feature	Gynecoid	Android	Anthropoid	Platypelloid
Inlet shape	**Rounded**	**Heart-shaped**	**Oval** (anteroposterior)	**Wide, flattened** oval
Cavity	Roomy	Funnel-shaped	Deep and narrow	Shallow
Subpubic angle	Wide	**Narrow**	Moderately wide	Wide
Sacrum	Broad, well curved	Inclined forward	Long and narrow	Short and broad
Labour	**Most favourable** for vaginal delivery	**Increased risk** of difficult labour	Vaginal delivery often possible	**Engagement** of the fetal head may be difficult

4 Walls of the pelvic cavity

Anterior	The **shallowest** wall · **symphysis pubis** (secondary cartilaginous = fibrocartilaginous joint, reinforced by superior and inferior ligaments) · **pubic rami**
Posterior	**Sacrum and coccyx** · **piriformis** · fascia
Lateral	**Hip bone** · obturator membrane · **obturator internus** and its fascia · **sacrotuberous and sacrospinous ligaments**
Floor	**Pelvic diaphragm** (section 5)



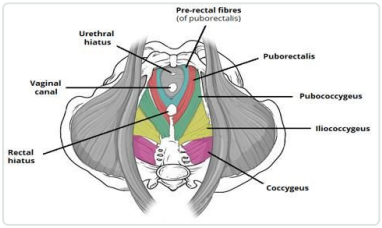
Piriformis (posterior) and obturator internus (lateral).

Muscle	Origin	Insertion	Nerve	Action
Piriformis posterior wall	Pelvic surface of **sacrum**	**Greater trochanter**	—	Lateral rotation of femur
Obturator internus lateral wall	Inner surface of **obturator membrane**	**Medial border** of greater trochanter	Nerve to obturator internus*	Lateral rotation of thigh

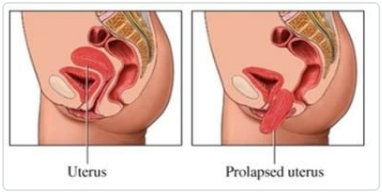
*The slide gives its roots as L4, S1. Many textbooks give L5–S2 — confirm with the lecturer before memorising the roots.

5 Pelvic floor (pelvic diaphragm)

What	**Funnel-shaped** structure separating the **pelvic cavity** above from the **perineum** below
Components	**Levator ani** (largest) · **coccygeus** · fascia covering both
Levator ani	Broad sheet of **three paired muscles**: **puborectalis**, **pubococcygeus**, **iliococcygeus**
Nerve	**Anterior ramus of S4** + branches of the **pudendal nerve** (S2–S4)
Gaps	**Urogenital hiatus** — anterior · urethra (+ vagina in females) · **rectal hiatus** — central · anal canal
Injury	Difficult childbirth can **tear the perineal body** → weak floor → **uterine prolapse** (and rectal prolapse)



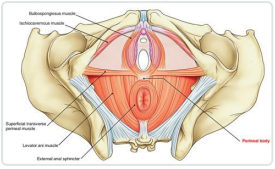
Levator ani parts, coccygeus, urethral and rectal hiatus.

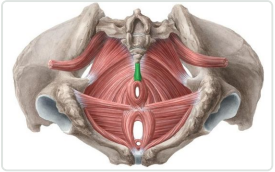


Normal vs prolapsed uterus.

6 Perineal body vs anococcygeal body

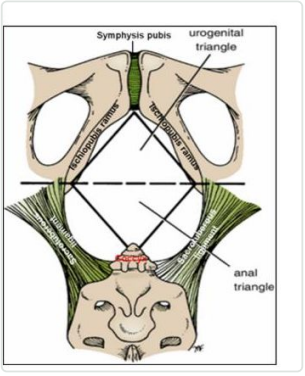
	Perineal body	Anococcygeal body
Nature	**Pyramidal fibromuscular** mass	Complex **musculotendinous** structure
Where	Middle of the junction of the **urogenital and anal triangles**	Between the **anterior aspect of the coccyx** and the **posterior wall of the anorectal canal**
Attached	**External anal sphincter** · **bulbospongiosus** · **superficial transverse perineal** · levator ani (anterior part)	Receives fibres of **levator ani**
Clinical	Torn in difficult childbirth → prolapse	—


Perineal body.


Anococcygeal body (green).

7 Perineum — boundaries and the two triangles

Shape	**Diamond-shaped** area between the thighs, **below the pelvic diaphragm**
Bony boundaries	Anterior: **symphysis pubis** · posterior: **coccyx** · lateral: **ischiopubic rami, ischial tuberosities, sacrotuberous ligaments**
Division	An imaginary line **through the two ischial tuberosities** → urogenital triangle in front, anal triangle behind
Contents	Lower ends of the **urethra, vagina and anal canal** · external genitalia · **perineal and anococcygeal bodies**



Line between the ischial tuberosities splits the diamond.

	Urogenital triangle	Anal triangle
Anterior	Symphysis pubis	Line through the 2 ischial tuberosities
Posterior	Line through the 2 ischial tuberosities	**Coccyx**
Lateral	Ischiopubic rami + ischial tuberosities	Ischial tuberosity + **sacrotuberous ligament**
Contents	Lower urethra + vagina · external genitalia (vulva)	Lower **anal canal** · anococcygeal body · **ischiorectal fossa** each side

8 Urogenital diaphragm and the perineal pouches

UG diaphragm	Triangular **musculofascial** diaphragm in the anterior perineum · fills the gap of the **pubic arch** · **sphincter urethrae** + **deep transverse perineal** muscles between superior and inferior fasciae
Perineal membrane	= the **inferior fascia of the UG diaphragm** · it **separates** the superficial and deep pouches
Superficial pouch	Between the **membranous layer of superficial fascia (Colles')** below and the **perineal membrane** above · laterally **ischiopubic rami**
Deep pouch	**Completely closed** space deep to the perineal membrane · above: **superior fascia of UG diaphragm** · laterally: inferior part of **obturator internus fascia**

PERINEAL LAYERS — what sits in each pouch



	MALE	FEMALE
Superior fascia of UG diaphragm	**Membranous (intermediate) urethra** External urethral sphincter Deep transverse perineal (variable) **Bulbourethral (Cowper's) glands** Dorsal n. + vessels of penis	Part of urethra + vagina **External urethral sphincter complex:** sphincter urethrae · compressor urethrae · urethrovaginal sphincter Deep transverse perineal (variable) Dorsal n. + vessels of clitoris
Perineal membrane = inferior fascia of UG diaphragm	Bulb + right and left crura of penis Bulbospongiosus · ischiocavernosus · superficial transverse perineal **Proximal spongy (bulbar) urethra** Perineal br. of int. pudendal vessels and of the pudendal nerve	Vestibular bulbs + clitoral crura Bulbospongiosus · ischiocavernosus · superficial transverse perineal **Greater vestibular (Bartholin) glands** Perineal br. of int. pudendal vessels and of the pudendal nerve
Colles' fascia (membranous layer of superficial fascia)		

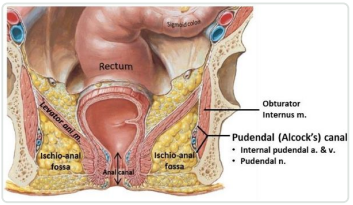
Bold = the item that tells the two pouches apart.

The look-alike glands: Bartholin's (greater vestibular) glands are in the **superficial** pouch of women · **Cowper's** (bulbourethral) glands are in the **deep** pouch of men. The **membranous** urethra is deep; the **bulbar** urethra is superficial.

Link to SGL 1 (urethral injury): a torn **bulbar** urethra bleeds into the superficial pouch and spreads deep to Colles' fascia into the scrotum and penis; a torn **membranous** urethra bleeds **above the perineal membrane**.

9 Anal triangle and ischiorectal fossa

Ischiorectal fossa	Fascia-lined, **wedge-shaped** space on **each side of the anal canal**
Base	**Skin** of the perineum
Medial wall	**Levator ani** and the **anal canal**
Lateral wall	**Obturator internus**, covered with fascia
Contents	**Dense fat** · **pudendal nerve + internal pudendal vessels** in the **pudendal (Alcock's) canal** · **inferior rectal nerve and vessels** crossing the fossa to the anal canal



Ischio-anal fossae and the pudendal (Alcock's) canal.

P Must-remember pearls

- Inlet: symphysis · **sacral promontory** + ala · **iliopectineal lines** · outlet: symphysis · **coccyx** · ischiopubic rami · **sacrotuberous ligs**.
- Gynecoid = rounded, **best for delivery** · android = heart-shaped, male-like · anthropoid **AP > transverse** · platypelloid **transverse > AP**.
- Pelvic diaphragm = **levator ani** (3 parts) + coccygeus · S4 + pudendal · torn **perineal body** → prolapse.
- Superficial pouch: erectile tissue, 3 muscles, **Bartholin's** · deep: membranous urethra, ext. sphincter, **Cowper's**.
- Ischiorectal fossa: medial **levator ani**, lateral **obturator internus** · **Alcock's canal** = pudendal n. + internal pudendal vessels.

Q Test yourself

1. Which structure joins the front of the coccyx to the back of the anorectal canal?

A. Perineal body
B. Anococcygeal body
C. Sacrospinous ligament
D. Perineal membrane
E. Coccygeus

2. A muscle runs from the pelvic surface of the sacrum to the greater trochanter. Name and action?

A. Obturator internus — laterally rotates the thigh
B. Piriformis — medially rotates the femur
C. Iliacus — laterally rotates the femur
D. Piriformis — laterally rotates the femur
E. Obturator internus — medially rotates the thigh

3. Rounded inlet, roomy cavity, wide subpubic angle — what does this pelvis predict?

A. Most favourable pelvis for vaginal delivery
B. Increased risk of difficult labour
C. Difficult engagement of the fetal head
D. AP diameter exceeding the transverse
E. A pelvis resembling the male type

4. Which structure passes through the rectal hiatus of the pelvic floor?

A. Urethra
B. Vagina
C. Pudendal nerve
D. Sciatic nerve
E. Anal canal

5. Which nerve supply of levator ani is given in the lecture?

A. Obturator nerve (L2–L4)
B. Pelvic splanchnic nerves only
C. Anterior ramus of S4 + branches of the pudendal nerve
D. Femoral nerve (L2–L4)
E. Inferior gluteal nerve

ANSWERS — COVER UNTIL YOU'VE COMMITTED

1 B — Anococcygeal body: coccyx ↔ anorectal canal; the perineal body lies at the junction of the two triangles.
2 D — Piriformis forms part of the posterior wall: sacrum → greater trochanter, lateral rotation of the femur.
3 A — These are gynecoid features — the classical female type and the most favourable for vaginal delivery.
4 E — Rectal hiatus (central) → anal canal; urogenital hiatus (anterior) → urethra ± vagina.
5 C — Levator ani: anterior ramus of S4 and branches of the pudendal nerve (roots S2, S3, S4).

SPACED REVIEW

☐ Day 1 | ☐ Day 3 | ☐ Day 7 | ☐ Day 21 | ☐ Pre-exam